

Field Oriented Control of a Seven-Phase Series Connected Three PMSM Using Fuzzy Logic

Abdesslam OUANOUI^{(1)*}, Katia KOUZI⁽¹⁾, DjamelDIFI⁽²⁾, Mohamed GHIBECHE⁽¹⁾,
And mouradSELLAH⁽³⁾

⁽¹⁾ Department of Electrical Engineering, Faculty of TechnologyLaghouat University, Algeria

⁽²⁾ National Polytechnic School Algiers,Algeria

⁽³⁾ Department of Electrical Engineering, Faculty of Technology djelfa University, Algeria

*ouanouki81@yahoo.fr

Abstract: The aim of this paper is to present field oriented -based fuzzy logic control seven-phase series connected three permanent magnet synchronous machines, fed by a single inverter and controlled independently. where a fuzzy logic -based speed controller has been constructed and used to drive the three-motors. The three-motor system, inverter system, and the fuzzy controller models are implemented and tested using MATLAB/SIMULINK facilities. The obtained results show that the FLC possesses a remarkable performance and a faster dynamic response compared to the traditional field-oriented control.

Keywords: seven-phase permanent magnet synchronous machines (PMSM) ; Fuzzy logic control; series-connected; three-motor; vector control.

1. INTRODUCTION

Polyphase motors have recently been studied extensively by researchers, due to the advantages they have compared to traditional motors, such as reduced torque ripple. They are also considered as solution to the high power problem[1-5], where the energy and currents are divided between a larger numbers of phases. Therefore, polyphase motors are used in Textile and paper industry, robotics, traction, electric vehicles, etc [6-8-9]. The new concept of multi-motor propulsion systems, based on the use of multi-phase machines, is the independent control of a group of motors connected in series and supplied from a single VSI. The PMSM with three seven-phase motors was analyzed [10-15]. The idea stems from the fact that any n-phase machine requires two independent flux and torque control currents. Thus, additional degrees of freedom exist in a polyphase machine, which can be used to control other machines . It has been proposed that by connecting the stator windings of the seven-phase series connected three machines with suitable phase switching., the independent beam control of a three-PMSM seven phases system connected in series which is supplied by a single seven-phase inverter is investigated in this paper[17-20].

To overcome the main disadvantages, the fuzzy logic controllers suitable for the speed control of series-connected seven-phase three PMSM has been presented [21-23].

The fuzzy logic controller (FLC) is not based on a mathematical model [24-28], is widely

used to solve problems under uncertain and vague environments, have been successfully applied in many control applications, especially for speed control of motors such as PMSM.

The purpose of this paper is to investigate independent Control of Series-Connected seven-phase three PMSM supplied from a single seven-leg inverter using fuzzy logic control.

2. MODELLING OF A SEVEN-PHASE SYNCHRONOUS MACHINE

The model of the seven-phase PMSM is expressed by the following equations [3]:

$$[v_{ABCDEFG}] = R_s [i_{ABCDEFG}] + \frac{d}{dt} [\phi_{ABCDEFG}](1)$$

The flux equation is given by:

$$[\phi_{abcdefg}] = [L_{s1}] [i_{abcdefg}] + [\phi_{aimant}](2)$$

Flux vector created by the magnet through the stator winding:

$$[\phi_{aimant}] = \begin{bmatrix} \phi_f \cos(\theta) \\ \phi_f \cos(\theta - (2\pi/7)) \\ \phi_f \cos(\theta - (4\pi/7)) \\ \phi_f \cos(\theta - (6\pi/7)) \\ \phi_f \cos(\theta - (8\pi/7)) \\ \phi_f \cos(\theta - (10\pi/7)) \\ \phi_f \cos(\theta - (12\pi/7)) \end{bmatrix} \quad (3)$$

3. MATHEMATICAL MODEL OF THREE -PMSM SEVEN PHASES SYSTEM

The stator windings of a seven-phase series connected three permanent magnetsynchronous machines as shown in Figure 1. All three machines are controlled independently by switching the appropriate phase. Therefore, when the first machine produces a rotating magnetic field, it does not produce flux and torque. In the second and third machines, likewise, when the second machine produces a rotating

magnetic field, the first and third machines do not produce any torque, and in the end, when the third machine produces flow and torque, the first and second machines will not produce any torque.

The three machines are powered by a single seven-phase inverter. The spatial displacement between any two successive phases in the machines is $\alpha = 2\pi / 7$. [1-5-6].

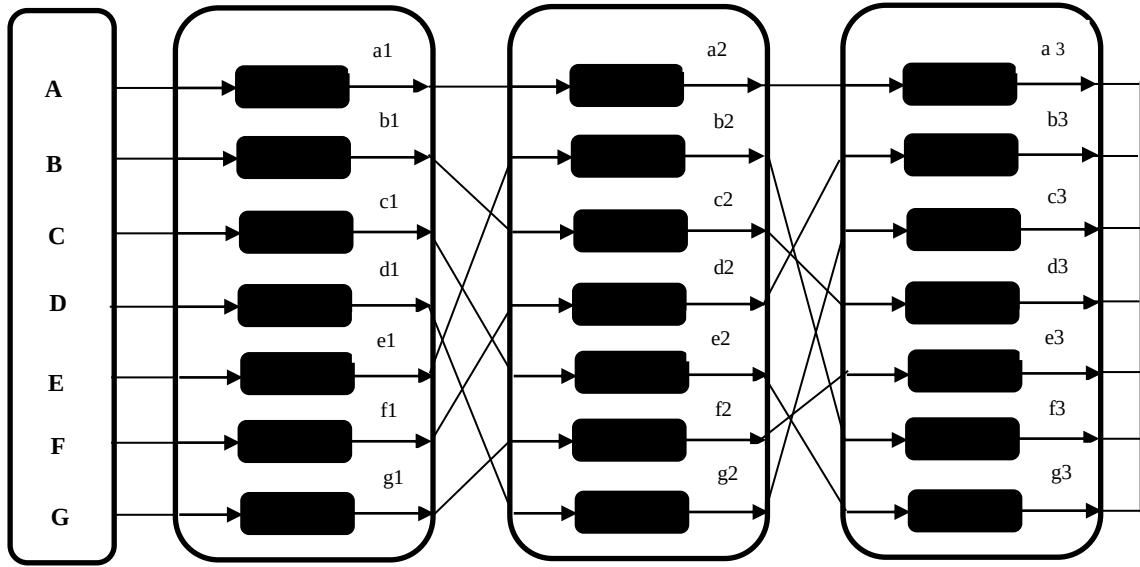


fig1. the diagram for Seven Phase Series Connected Three synchronous Motor system

The voltages are determined by the connection diagram of the stator windings of the machines shown in Figure 1 according to [2-6]:

$$\begin{cases} v_A = v_{as1} + v_{as2} + v_{as3} \\ v_B = v_{bs1} + v_{cs2} + v_{ds3} \\ v_C = v_{cs1} + v_{es2} + v_{gs3} \\ v_D = v_{ds1} + v_{gs2} + v_{cs3} \\ v_E = v_{es1} + v_{bs2} + v_{fs3} \\ v_F = v_{fs1} + v_{ds2} + v_{bs3} \\ v_G = v_{gs1} + v_{fs2} + v_{es3} \end{cases} \quad (4)$$

the relationship between the output currents of the inverter and the phase currents of three seven-phase machines are given with [3]:

$$\begin{cases} i_A = i_{as1} = i_{as2} = i_{as3} \\ i_B = i_{bs1} = i_{cs2} = i_{ds3} \\ i_C = i_{cs1} = i_{es2} = i_{gs3} \\ i_D = i_{ds1} = i_{gs2} = i_{cs3} \\ i_E = i_{es1} = i_{bs2} = i_{fs3} \\ i_F = i_{fs1} = i_{ds2} = i_{bs3} \\ i_G = i_{gs1} = i_{fs2} = i_{es3} \end{cases} \quad (5)$$

The Clark transformation matrix $[T_7(\alpha)]$ for a seven-phase system can be written as follows:

$$[T] = \begin{bmatrix} 1 & \cos \alpha & \cos 2\alpha & \cos 3\alpha & \cos 4\alpha & \cos 5\alpha & \cos 6\alpha \\ 0 & \sin \alpha & \sin 2\alpha & \sin 3\alpha & \sin 4\alpha & \sin 5\alpha & \sin 6\alpha \\ 1 & \cos 2\alpha & \cos 4\alpha & \cos 6\alpha & \cos 8\alpha & \cos 10\alpha & \cos 12\alpha \\ 0 & \sin 2\alpha & \sin 4\alpha & \sin 6\alpha & \sin 8\alpha & \sin 10\alpha & \sin 12\alpha \\ 1 & \cos 3\alpha & \cos 6\alpha & \cos 9\alpha & \cos 12\alpha & \cos 15\alpha & \cos 18\alpha \\ 0 & \sin 3\alpha & \sin 6\alpha & \sin 9\alpha & \sin 12\alpha & \sin 15\alpha & \sin 18\alpha \\ 1/\sqrt{2} & 1/\sqrt{2} & 1/\sqrt{2} & 1/\sqrt{2} & 1/\sqrt{2} & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \quad (6)$$

By using this matrix for relation (4) we will find the voltages of each machine in this reference:

$$\begin{bmatrix} v_{\alpha}^{INV} \\ v_{\beta}^{INV} \\ v_{x1}^{INV} \\ v_{y1}^{INV} \\ v_{x2}^{INV} \\ v_{y2}^{INV} \\ v_o^{INV} \end{bmatrix} = [T] \begin{bmatrix} v_{as1} + v_{as2} + v_{as3} \\ v_{bs1} + v_{bs2} + v_{bs3} \\ v_{cs1} + v_{cs2} + v_{cs3} \\ v_{ds1} + v_{ds2} + v_{ds3} \\ v_{es1} + v_{es2} + v_{es3} \\ v_{fs1} + v_{fs2} + v_{fs3} \\ v_{gs1} + v_{gs2} + v_{gs3} \end{bmatrix} \quad (7)$$

The corresponding mathematical model for the three synchronous machines connected in series in Fig1 in the fixed reference frame is given by the following equations [8]:

$$\begin{cases} v_{\alpha}^{INV} = R_s i_{\alpha}^{INV} + \left(l_{s1} + \frac{7}{2} m_{s1} + l_{s2} \right) \frac{d}{dt} i_{\alpha}^{INV} \\ \quad + l_{s3} \frac{d}{dt} i_{\alpha}^{INV} - \sqrt{\frac{7}{2}} \omega_1 \phi_{f1} \sin(\theta_1) \\ v_{\beta}^{INV} = R_s i_{\beta}^{INV} + \left(l_{s1} + \frac{7}{2} m_{s1} + l_{s2} \right) \frac{d}{dt} i_{\beta}^{INV} \\ \quad + l_{s3} \frac{d}{dt} i_{\beta}^{INV} + \sqrt{\frac{7}{2}} \omega_1 \phi_{f1} \cos(\theta_1) \\ v_{x1}^{INV} = R_{s1} i_{x1}^{INV} + \left(l_{s1} + l_{s2} + \frac{7}{2} m_{s2} \right) \frac{d}{dt} i_{x1}^{INV} \\ \quad + \left(l_{s3} + \frac{7}{2} m_{s3} \right) \frac{d}{dt} i_{x1}^{INV} - \sqrt{\frac{7}{2}} \omega_2 \phi_{f2} \sin(\theta_2) \\ v_{y1}^{INV} = R_s i_{y1}^{INV} + \left(l_{s1} + l_{s2} + \frac{7}{2} m_{s2} \right) \frac{d}{dt} i_{y1}^{INV} \\ \quad + \left(l_{s3} + \frac{7}{2} m_{s3} \right) \frac{d}{dt} i_{y1}^{INV} + \sqrt{\frac{7}{2}} \omega_2 \phi_{f2} \cos(\theta_2) \\ v_{x2}^{INV} = R_s i_{x2}^{INV} + \left(l_{s2} + \frac{7}{2} m_{s2} \right) \frac{d}{dt} i_{x2}^{INV} \\ \quad + \left(l_{s3} + \frac{7}{2} m_{s3} \right) \frac{d}{dt} i_{x2}^{INV} - \sqrt{\frac{7}{2}} \omega_3 \phi_{f3} \sin(\theta_3) \\ v_{y2}^{INV} = R_s i_{y2}^{INV} + \left(l_{s1} + l_{s2} + \frac{7}{2} m_{s2} \right) \frac{d}{dt} i_{y2}^{INV} \\ \quad + \left(l_{s3} + \frac{7}{2} m_{s3} \right) \frac{d}{dt} i_{y2}^{INV} + \sqrt{\frac{7}{2}} \omega_3 \phi_{f3} \cos(\theta_3) \end{cases} \quad (8)$$

With: $R_s = R_{s1} + R_{s2} + R_{s3}$

The torque relationships of the three machines connected in series are given in terms of the current components of the inverter by:

$$\begin{cases} T_{e1} = p \left((L_d - L_q) i_d^{INV} i_q^{INV} + \sqrt{\frac{7}{2}} \phi_{f1} i_q^{INV} \right) \\ T_{e2} = p \left((L_{x1} - L_{y1}) i_{x1}^{INV} i_{y1}^{INV} + \sqrt{\frac{7}{2}} \phi_{f2} i_{y1}^{INV} \right) \\ T_{e3} = p \left((L_{x2} - L_{y2}) i_{x2}^{INV} i_{y2}^{INV} + \sqrt{\frac{7}{2}} \phi_{f3} i_{y2}^{INV} \right) \end{cases} \quad (9)$$

4. THE INDEPENDENT VECTOR CONTROL OF SERIESCONNECTED THREE - PMSM SEVEN PHASES

The objective is to accomplish the independent control of all the polyphase machines in the group while using a single power inverter. The stator windings of the polyphase machines can be connected in series thus allowing the independent vector control of each machine, although the multi-motor drive complete system is fed from a polyphase power inverter

The three equations are completely independent, so we can control each machine with three vector controls and using a single power inverter[1-6-7].

$$\begin{cases} v_d^{INV} = \left(R_s + P \left(l_{s1} + \frac{7}{2} m_{s1} + l_{s2} + l_{s3} \right) \right) i_d^{INV} \\ \quad - \omega_1 \left(l_{s1} + \frac{7}{2} m_{s1} \right) i_q^{INV} \\ v_q^{INV} = \left(R_s + P \left(l_{s1} + \frac{7}{2} m_{s1} + l_{s2} + l_{s3} \right) \right) i_q^{INV} \\ \quad + \omega_1 \left(l_{s1} + \frac{7}{2} m_{s1} \right) i_d^{INV} + \sqrt{\frac{7}{2}} \omega_1 \phi_{f1} \end{cases} \quad (10)$$

The second machine:

$$\begin{cases} v_{x1}^{INV} = \left(R_s + P \left(l_{s2} + \frac{7}{2} m_{s2} + l_{s1} + l_{s3} + \frac{7}{2} m_{s3} \right) \right) i_{x1}^{INV} \\ \quad - \omega_2 \left(l_{s2} + \frac{7}{2} m_{s2} \right) i_{y1}^{INV} \\ v_{y1}^{INV} = \left(R_s + P \left(l_{s2} + \frac{7}{2} m_{s2} + l_{s1} + l_{s3} + \frac{7}{2} m_{s3} \right) \right) i_{y1}^{INV} \\ \quad + \omega_2 \left(l_{s2} + \frac{7}{2} m_{s2} \right) i_{x1}^{INV} + \sqrt{\frac{7}{2}} \omega_2 \phi_{f2} \end{cases} \quad (11)$$

The third machine:

$$\begin{cases} v_{x2}^{INV} = (R_s + P(l_{s2} + m_{s2} + l_{s1} + l_{s3} + m_{s3}))i_{x2}^{INV} \\ \quad - \omega_3(l_{s3} + \frac{7}{2}m_{s3})i_{y2}^{INV} \\ v_{y2}^{INV} = (R_s + P(l_{s2} + m_{s2} + l_{s1} + l_{s3} + m_{s3}))i_{y2}^{INV} \\ \quad + \omega_3(l_{s3} + m_{s3})i_{x2}^{INV} + \sqrt{\frac{7}{2}}\omega_3\phi_{f2} \end{cases} \quad (12)$$

The first machine torque controlled by the two currents (I_d^{INV} , I_q^{INV}); for the second machine the torque controlled by the two currents (I_{x1}^{INV} , I_{y1}^{INV}). And for the third machine the torque controlled by the two currents (I_{x2}^{INV} , I_{y2}^{INV}). Among the control strategies, we often use the one which consists in maintaining the component $I_d^{INV}I_{x1}^{INV}$ and I_{x2}^{INV} zero. We control the torques only by the currents I_q^{INV} ; I_{y1}^{INV} . and I_{y2}^{INV} . The speeds are thus regulated by the component I_q^{INV} and I_y^{INV} [5-7].

5. CONTROL BY THE FUZZY LOGIC

The fuzzy control system approaches the flexibility of human reasoning to some level; he achieves excellent performance without the requirement for a mathematical model of the system, just by incorporating experts' knowledge. Fuzzy control, in general, is a model-free strategy that does not rely on the model of the system under control. Because obtaining a mathematical model of the system can be a difficult process, model-free approaches make controller design easier [21-25].

Fuzzification, fuzzy inference engine, rule base, and defuzzification are the four essential components of a fuzzy logic controller, as shown in Figure. Inputs are fuzzified, then outputs are generated using a rule basis and inference system, and the fuzzy outputs are defuzzified and applied to the primary control system. Error of inputs from their references and error deviations in any time interval are chosen as inputs. Mamdani

type fuzzy logic control is considered here [26-28].

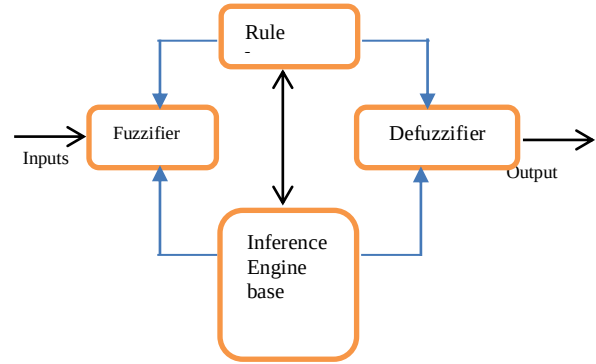


Fig 2. Structure of fuzzy logic controller

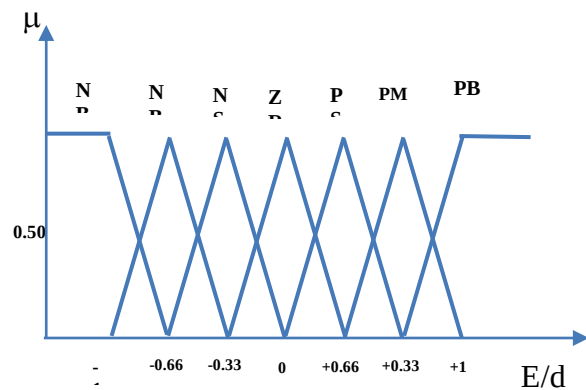


Fig 3. Membership functions of input/output variables: speed and current.

In this study, seven terms are assigned in Table I: NL, negative large; NM, negative medium; NS, negative small; ZE, zero; PS, positive small; PM, positive medium; and PL, positive large. Each fuzzy variable is a member of the subsets with a degree of membership μ varying between 0 and 1 (Figure 3). The rules have been written in matrix (Table I). Each rule is expressed in the form as:

- R1 : If E is NL and dE is PL then μ is ZR
- R2 : If E is NM and dE is PL then μ is PS
- R49 : If E is PL and dE is NL then μ is ZR

Table 1. The rule base for controlling the speed and the current

dE	E						
	NL	NM	NS	ZR	PS	PM	PL
PL	ZR	PS	PM	PL	PL	PL	PL
PM	NS	ZR	PS	PM	PL	PL	PL
PS	NM	NS	ZR	PS	PM	PL	PL
ZE	NL	NM	NS	ZR	PS	PM	PS
NS	NL	NL	NM	NS	ZR	PS	PM
NM	NL	NL	NL	NM	NS	ZR	PS
NL	NL	NL	NL	NL	NM	NS	ZR

The same steps used for the conception of the speed controller will be repeated for the currents controller, only we have:

The first machine:

Input error E : instead of being equal to $E = \Omega_{1ref} - \Omega_1$, it will be equal with $E = i_{dref} - i_d$ for the first fuzzy controller of current i_{ds} and $E = i_{qref} - i_q$ for the second fuzzy controller of current i_{qs} ;

The output of the fuzzy controller is V_{ds} for the i_{ds} current controller and V_{qs} for the i_{qs} current controller.

The second machine:

Input error E : instead of being equal to $E = \Omega_{2ref} - \Omega_2$, it will be equal with $E = i_{x1ref} - i_{x1}$ for the first fuzzy controller of current i_{ds} and $E = i_{y1ref} - i_{y1}$ for the second fuzzy controller of current i_{y1s} ;

The output of the fuzzy controller is V_{x1s} for the i_{x1s} current controller and V_{y1s} for the i_{y1s} current controller.

The third machine:

Input error E : instead of being equal to $E = \Omega_{3ref} - \Omega_3$, it will be equal with $E = i_{x2ref} - i_{x2}$ for the first fuzzy controller of current i_{y2s} and $E = i_{y2ref} - i_{y2}$ for the second fuzzy controller of current i_{ys} ;

The output of the fuzzy controller is V_{x2s} for the i_{x2s} current controller and V_{y2s} for the i_{y2s} current controller.

6. SIMULATION RESULTS

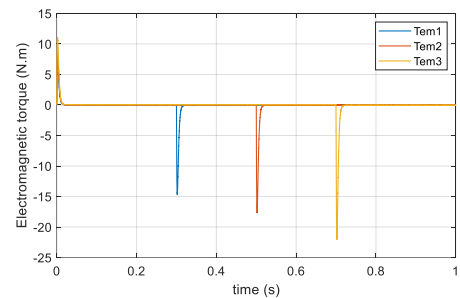
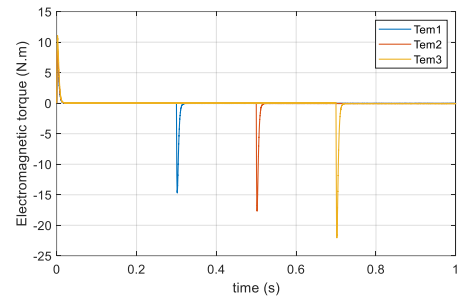
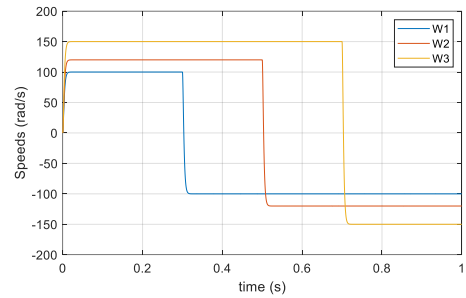
The fuzzy logic control of a seven-phase series connected three PMSM supplied by a Single inverter was simulated by using MATLAB/Simulink software.

The speeds responses of the motors are shown in Figure 4 under the no-load condition where the step in the reference speed of the

first machine was changed from 100 rad/s to -100 rad/s at $t = 0.3$ s. and the second machine from 120 rad/s to -120 rad/s at $t = 0.5$ s. And the third machine from 150 rad/s to -150 rad/s at $t = 0.7$ s.

Figure 5 shows the response of the motors under a variable step loading torque of up to 5 N.m at $t = 0.3$ s. For the first motor, 4 N.m at $t = 0.5$ s. For the second motor and 7 N.m at $t = 0.7$ sec for the third motor.

It is also noted that the current value of the three machines increases during the starting phase and then decreases to reach a fixed value. In the steady state and after applying the load torque, the current (I_q^{INV} , I_{y1}^{INV} and I_{y2}^{INV}) is proportional to the electromagnetic torque (T_{e1} , T_{e2} and T_{e3}). On the other hand, the currents (I_d^{INV} , I_{x1}^{INV} and I_{x2}^{INV}) remain equal to zero and follows the set point during the entire operating cycle.



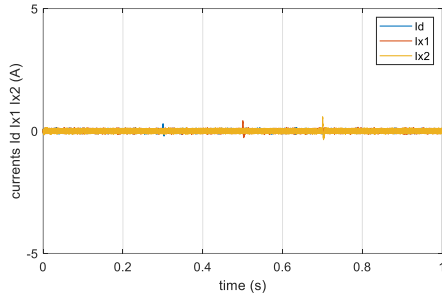


Fig 4. Simulation results of the Fuzzy logic control of a seven-phase series connected three PMSM (W1 +100 to -100, W2 +120 to -120; and W3 150 to -150)

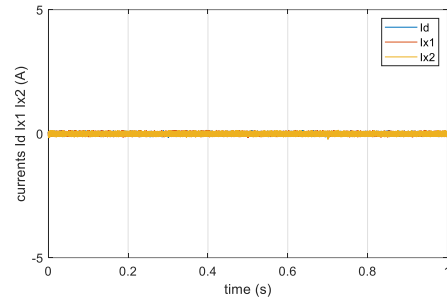
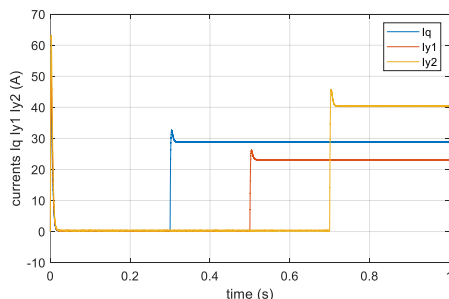
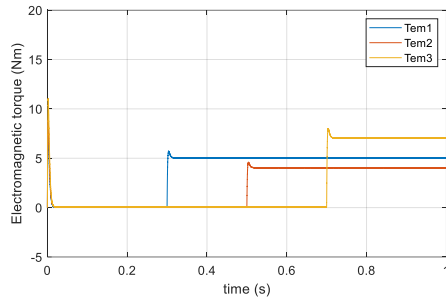
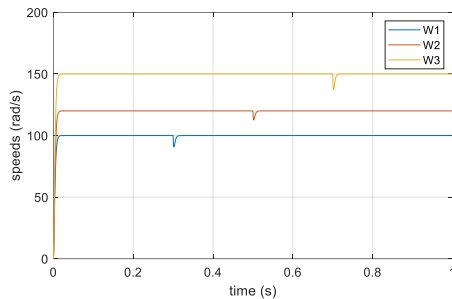


Fig 5. Simulation results of the Fuzzy logic control of a seven-phase series connected three PMSM under variable load torque test



7. CONCLUSIONS

In this paper, model of the three PMSM seven-phase connected in series is developed, which is supplied by single inverter seven phase. An FLC controller is employed, which can control the torque and current of each motor independently, the simulation results shows that the suggested controller in this study is capable of tracking the speed and current references, as well as being robust against rapid changes in load.

References

- [1] Moidu Thavot, Atif Iqbal, Mohammad Saleh, Akhtar Kalam " Modeling of A Seven-Phase Series-connected Three Motor Drive System" 2013 IEEE GCC Conference and exhibition, November 17-20, Doha, Qatar.
- [2] M. Jones, E. Levi, S.N. Vukosavic and H.A. Toliyat "Independent Vector Control of a Seven-Phase Three-Motor Drive System Supplied from a Single Voltage Source Inverter" IEEE 34th Annual Conference on Power Electronics Specialist, 2003. PESC '03.
- [3] E. Levi, M. Jones, S.N. Vukoasavic, H.A. Toliyat, "A novel nine-phase four-motor drive system with completely decoupled control" Proc. 29th IEEE IECON, vol. 1, pp.637-642, 2003.
- [4] ZHANG Hua-qiang, LU Xiao-tong, WANG Xin-sheng, WANG Xue-yi and LENG Yan-li "Space Vector PWM for Three Series-Connected Permanent Magnet Synchronous Motors Drive System" International Power, Electronics and Materials Engineering Conference (IPEMEC 2015)
- [5] Kamel Saleh "Seven-Phase PMSM Drives Operation Post Two Types of Faults" MDPI Appl. Sci. 2022, 12, 7979. <https://doi.org/10.3390/app12167979>
- [6] E. Levi, A. Iqbal, S. N. Vukosavic, and H. A. Toliyat, "Modeling and control of a five-phase series-connected two-motor drive," Conference on IEEE Industrial Electronics Society (IECON'03), 2003, pp. 208–213

- [7] M. Jones, E. Levi, and A. Iqbal, "A five-phase series-connected twomotor drive with current control in the rotating frame," *Power Electronics Specialists Conference (PESC'04)*, 2004, pp. 3278–3284.
- [8] H. C. Chen and C. X. Su, "Current control for single-inverter-fed series-connected five-phase PMSMs," in *Proc. IEEE ISIE*, Taipei, Taiwan, 2013, pp. 1-6.
- [9] Navid R. Abjadi, "Sliding-mode control of a six-phase series/parallel connected two induction motors drive," *ISA Trans.*, vol. 53, no. 6, pp. 1847-1856, Nov. 2015
- [10] Djamel DIFI, Khaled HALBAOUI, Djamel BOUKHETALA, Hybrid control of five-phase permanent magnet synchronous machine using space vector modulation *Turkish Journal of Electrical Engineering & Computer Sciences* (2019) 921 – 938
- [11] M. R. Khan and A. Iqbal, "MRAS Based Sensorless Control of a Series-Connected Five-Phase Two-Motor Drive System," *Journal of Electrical Engineering & Technology*, vol. 3, no. 2, pp. 224-234, 2008.
- [12] R. Omata, K. Oka, A. Furuya, S. Matsumoto, Y. Nozawa and K. Matsuse, "An improved performance of five-leg inverter in two induction motor drives," in *Proc. Int. Power Electronics and Motion Control Conf. IPEMC*, pp. 1-5, 2006.
- [13] Monmasson. E, Louis. J.P. Presentation of a control law for IM drive based on the dynamic reconfiguration of a DTC algorithm and a SVM-DTC algorithm. *Mathematics and Computers in Simulation*. 2003; 63: 321–333.
- [14] H. H. Chen and C.-X. Su, "Current Control for Single-Inverter Fed Series-Connected Five-Phase PMSMs," *IEEE International Symposium on Industrial Electronics*, Taipei, Taiwan, pp.1-6, 28- 31May, 2013
- [15] Sandre-Hernandez . O, Rangel-Magdaleno. J, MoralesCaporal. R, Bonilla-Huerta. E. HIL simulation of the DTC for a three-level inverter fed a PMSM with neutral-point balancing control based on FPGA. *Electrical Engineering*.September 2018, Volume 100, Issue 3, pp 1441–1454.
- [16] Kamel Salehand Mark Sumner "Sensorless Control of Seven-Phase PMSM Drives Using NSV-SVPWM with Minimum Current Distortion" *MDPI Electronics* 2022, 11(5), 792.
- [17] Khedher. A, Mimouni. M.F. Sensorless-adaptive DTC of double star induction motor. *Energy Conversion and Management* 2010; 51: 2878–2892.
- [18] Ammar. A, Benakcha. A, Bourek. A. Closed loop torque SVM-DTC based on robust super twisting speed controller for induction motor drive with efficiency optimization. *international journal of hydrogen energy* 2017; xxx: 1-3
- [19] A. Hosseini, R. Trabelsi, A. Iqbal, and M. F. Mimouni, "Backstepping Control for a Five-Phase Permanent Magnet Synchronous Motor Drive," *International Journal of Power Electronics and Drive System(IJPEDS)*, vol. 6, no. 4, pp. 842-852, December 2015.
- [20] Pimkumwong. N, Wang. M.S.Full-order observer for direct torque control of induction motor based on constant V/F control technique. *ISA Transactions* 2018; 73: 189-200.
- [21] Beghdadi M, Kouzi . A novel sensorless fully synergetic control of the brushless doubly-fed induction machine integrated in a wind-energy conversion system with a fuzzy-based HCS MPPT algorithm and an extended Kalman filter. *ELEKTROTEHNIŠKI VESTNIK* 86(4): 169-180, 2022
- [22] Islam Benhamida, Aissa Ameur, Katia Kouzi, Bachir Gaoui Torque Ripple Minimization in Predictive Torque Control Method of PMSM Drive Using Adaptive Fuzzy Logic Modulator and EKF Estimator *Journal of Control, Automation and Electrical Systems* <https://doi.org/10.1007/s40313-019-00505-7>
- [23] Bessaad Taieb, Rachid Taleb, Fayçal Chabni, and Atif Iqbal (2019); Fuzzy adaptive control of a multimachine system with single inverter supply. *International Transactions on Electrical Energy Systems*<https://doi.org/10.1002/2050-7038.12070>.
- [24] Listwan, J.; Pienkowski, K. Comparison of DFOC of Seven-Phase Induction Motor with PI and Fuzzy-Logic speed Controller under speed sensor fault. *Przeegl ,ad Elektrotechniczny* 2020, 96, 9–13
- [25] M. Badr, Ali M.Eltamaly, A.I.Alolah, Fuzzy Controller for Three Phase Induction Motor Drives by Basem, *International Autonomous and Intelligent Systems*, Portugal, 21–23 June 2010.
- [26] T. Hu and X. Zhang, "Simulation of PMSM vector control system based on fuzzy PI controller," in 2019 *IEEE International Conference on Power, Intelligent Computing and Systems (ICPICS)*, Jul. 2019, pp. 111–114, doi:10.1109/ICPICS47731.2019.8942439
- [27] K. Viji, A. Kumar, and R. Nagaraj, "Improved delta operator based discrete sliding mode fuzzy controller for buck converter," *Indian Journal of Science and Technology*, vol. 10, no. 25, pp. 1–11, Feb. 2017, doi: 10.17485/ijst/2017/v10i25/116507
- [28] G. Shahgholian, M. Maghsoodi, A. Movahedi, Fuzzy and Proportional Integral Controller Design for Thyristor Controlled Series Capacitor and Power System Stabilizer to Improve Power System Stability, *Rev. Roum. Sci. Techn. – Électrotechn. et Énerg.*, 61, 4, pp. 418–423 (2016)